

Time Series Analysis & Forecasting

Data $\rightarrow$ Information

Define a problem $\rightarrow$ Collect Data

Data $\rightarrow$ Quantitative and Qualitative

Types of Data

$\rightarrow$ Time Series Data : Data collected over a period of time and arranged in chronological order with equal time interval.

$\rightarrow$ Cross Sectional Data : Data collected at a point of time across different subjects

$\rightarrow$ Panel Data : Combination of timeseries & cross sectional

Balanced $\rightarrow$ same interval

Unbalanced $\rightarrow$ unequal interval

Forecasting

short term $\rightarrow$ 1-3 years months

medium term $\rightarrow$ 3-6 years months

long term $\rightarrow$ 6 - years more months

Univariate Data
 $\rightarrow$ sales.

$$y = C + S + t + i$$

→ Cyclical
 $> 1y$

→ Seasonality
 $< 1y$

→ Trend

→ Irregular
 cost price

Prediction and Forecasting

Forecasting based on past and present data

Prediction doesn't have a base

Additive Model

Multiplicative Model

$$y = I + t + S + C \rightarrow \text{short}$$

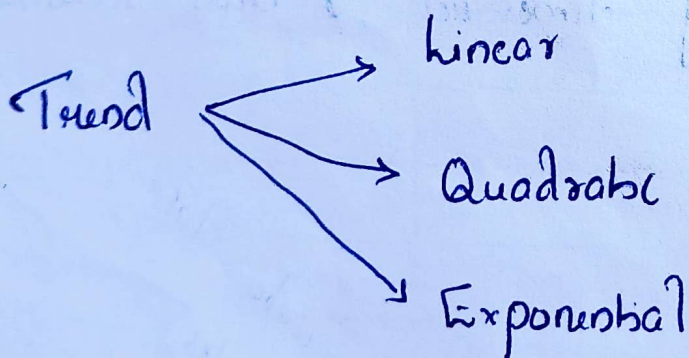
not have much trend

should convert into

$$y = I \times t \times S \times C$$

long with trend

additive model using log

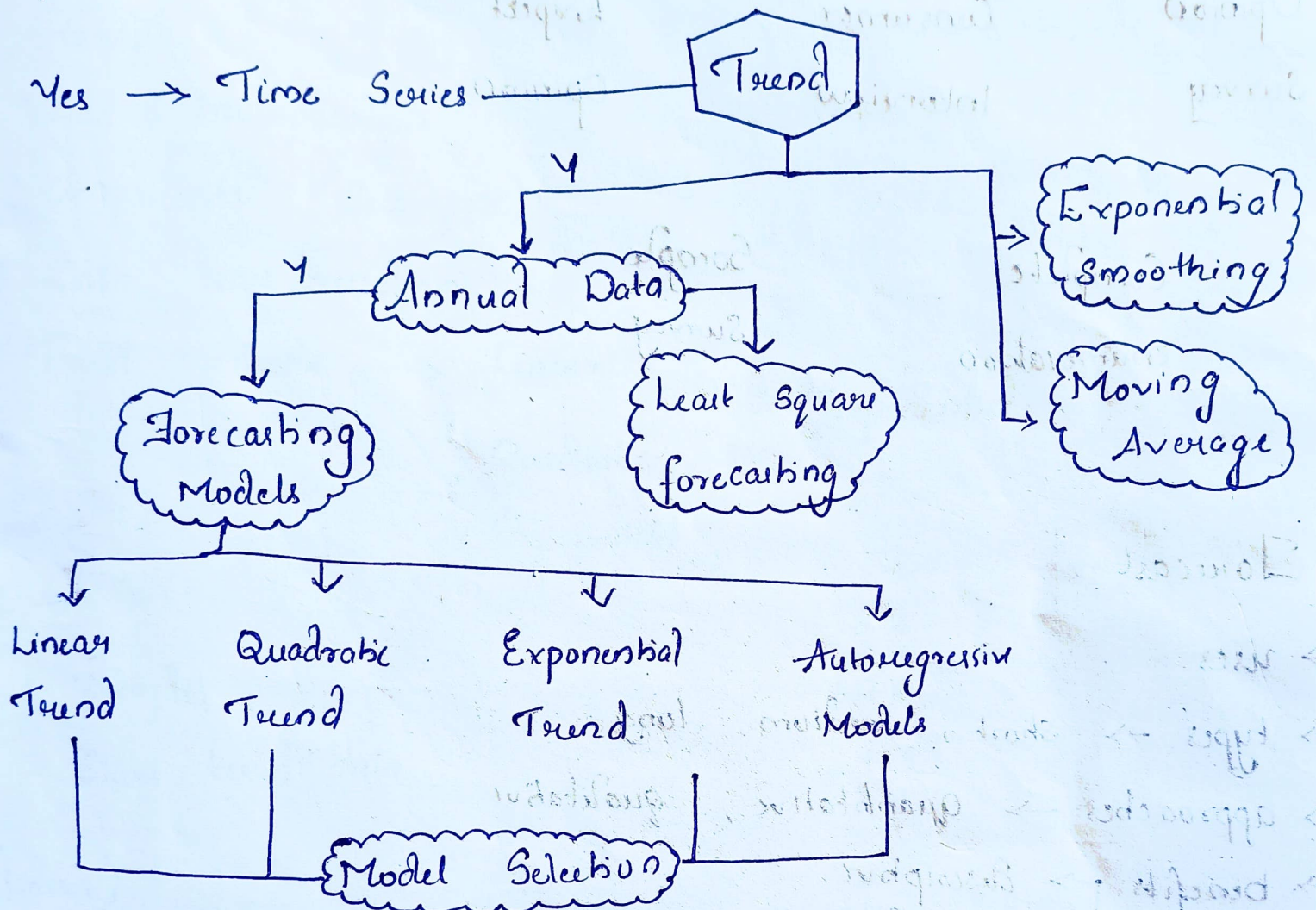


usually using multiplicative model but application is hard.

Forecasting Method

1) Quantitative

Yes → Time Series



No → Casual [Regression and Correlation] method

2) Qualitative $\rightarrow$ Survey Methods

Opinion
Survey

Consumer
Interview

Expert
Opinion

Complete
enumeration

Sample
Survey

Forecast

$\rightarrow$ uses

$\rightarrow$ types $\rightarrow$ short, medium, long

$\rightarrow$ approaches $\rightarrow$ Quantitative, qualitative

$\rightarrow$ benefits $\left\{ \begin{array}{l} \rightarrow$ descriptive
 $\rightarrow$ spatial
 $\rightarrow$ explorative \end{array} \right.

T $\rightarrow$ General measurement

S $\rightarrow$ Regular & recurring

C $\rightarrow$ long term flow annual the terms

U-1

→ TS

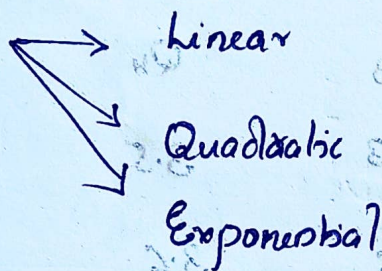
→ Graph

→ Add & Multiplication

→ Components (decompose)

→ Data transformation

→ Trend methods



(OLS)

1. Simple

2. Short term data

(Linear).

1. Assumes constant slope

2. Positive Data

Exp

1. Financial Analysis

2. Compound % growth rate

1. e^x

2. +ve data

Linear - $a + bx = y$

2020 — 22680

2021 — 22400

2022 — 22500

$y = 22860 - 235t$

Exponential - $y = ae^{bt}$

y

t

8.2

1

9.4

2

11.8

3

14.2

:

$y = 3.819e^{0.3893t}$

Quadratic - $y = a + bt + ct^2$

y

t

1081

1

1062

2

1032

:

$y = 1119.362t + 2.37t^2$

Quad

1. Turning Pt data
2. non linear

1. Complete
2. Extra plotted too far

y	Q ₁	Q ₂	Q ₃	Q ₄
Q1	3.7	4.1	3.3	3.5
Q2	3.7	3.9	3.6	3.6
Q3	4.0	4.1	3.3	3.1
Q4	3.3	4.4	4.0	4.0
Q _{ni}	<u>3.615</u>	<u>4.215</u>	<u>3.55</u>	<u>3.55</u>

Total Average
 $O_n = 3.7625$

$$ST = \frac{Q_{A1}}{O_n} \times 100$$

$$Q_1 = \underline{\underline{91.6}}$$

$$113.6$$

$$\underline{\underline{94.3}}$$

$$\underline{\underline{94.4}}$$

Y_t	Y_{t-1}	Y_{t-2}	ΔY_t	Growth R
10			2	20
12	10		3	25
15	12	10	4	26.67
19	15	12	6	31.51
25	19	15	3	12
28	25	19	4	14.28
32	28	25		

$$\Delta^2 Y_t = (\Delta Y_t - \Delta Y_{t-1})$$

Times Series

Data collected over a period of time and arranged in chronological order with equal interval of time.

eg: Monthly inflation, Quarterly GDP, Yearly sales.

Objectives

- Understanding past behaviour
- Identify components
- Forecasting
- Understanding relationships
- Decision - making

TS $\rightarrow$ scope, uses, methods

Data transformation

Estimation - T. E. P

Smoothing Technique $\rightarrow$ MA
Exponential Smoothing method

Moving Average

$\rightarrow$ Simple moving average

$\rightarrow$ Double moving average

$\rightarrow$ Central moving average

$\rightarrow$ Weighted moving average

1. Simple moving average

a

b

c

d

e

$$A_1 = \frac{a+b+c}{3}$$

$$A_2 = \frac{b+c+d}{3}$$

$$A_3 = \frac{c+d+e}{3}$$

3y moving average

2. Double Moving Average

a -

b $A = \frac{A_1 + A_2 + A_3}{3}$

c $B = \frac{A_2 + A_3 + A_4}{3}$

d $C = \frac{A_3 + A_4 + A_5}{3}$

e -

3. Central Moving Average

Centre of simple moving average

a -

b A_1

$$\frac{A_1 + A_2}{2}$$

c A_2

d A_3 $\frac{A_2 + A_3}{2}$

e -

4. Weighted Moving Average

weights α, β, γ

a -

b $\frac{\alpha A + \beta B + \gamma C}{3}$

c $\frac{\alpha B + \beta C + \gamma D}{3}$

d -

Q. 80 84 80 88 98 92 84 88 80 100 84
96 92

Ans.	MA ₃	MOBILE	Central	Wighted
80	-			-
84	81.3		82.67	26.93
80	84	84.67	86.33	21.867
88	88.67	88.45	90.67	21.8
98	92.67	90.89	92	30.13
92	91.3	90.67	89.67	31.8
84	88	87.78	86	30
88	84	87.11	86.67	28.13
80	89.3	87.11	88.67	29.2
100	88	90.22	90.67	28.13
84	93.3	90.67	92	32.13
96	90.67			29.067
92	-			-

MDr	MDr	MA1
80	-	-
84	-	-
80	86	
88	88.4	86.57
98	88.4	87.71
92	90	87.14
84	88.4	90
88	88.8	89.42
80	87.2	89.14
100	89.6	89.14
84	90.4	-
96		
92		

Exponential Smoothing

$$F_{t+1} = Y_t + \alpha (A_t - F_t) \quad \alpha = 0.05$$

Sales	20	21	25	22	18	21	26	19	16
Ac	28	25	26	27	23	21			

$A_t \rightarrow$ actual

$F_t \rightarrow$ forecast current

$f_{t+1} \rightarrow$ future forecast

$$F_{t+1} = F_t + \alpha (A_t - F_t)$$

$$\begin{aligned}
 20 &\rightarrow 20 + 0.05 (20 - 20) = 20 \\
 21 &\rightarrow 20 + 0.05 (20 - 21) = 26.65 \\
 25 &\rightarrow 26.65 + 0.05 (26.65 - 25) = 25.08 \\
 22 &\rightarrow 25.08 + 0.05 (25.08 - 22) = 22.15 \\
 18 &\rightarrow 22.15 + 0.05 (22.15 - 18) = 18.20 \\
 21 &\rightarrow 18.20 + 0.05 (18.20 - 21) = 20.86
 \end{aligned}$$

$$\alpha = 0.05$$

$$F_1 = 20$$

$$F_2 \rightarrow 20 + 0.05 (20 - 20) = 20$$

$$F_3 \rightarrow 20 + 0.05 (21 - 20) = 20.35$$

$$F_4 \rightarrow 20.35 + 0.05 (25 - 20.35) = 20.58$$

$$F_5 \rightarrow 20.58 + 0.05 (22 - 20.58) = 20.65$$

$$F_6 \rightarrow 20.65 + 0.05 (18 - 20.65) = 20.52$$

$$F_7 \rightarrow 20.52 + 0.05 (21 - 20.52) = 20.54$$

$$F_8 \rightarrow 20.54 + 0.05 (26 - 20.54) = 20.819$$

$$F_9 = 20.819 + 0.05 (19 - 20.819) = 20.72$$

$$F_{10} = 20.72 + 0.05 (16 - 20.72) = 20.49$$

$$F_{11} = 20.49 + 0.05 (28 - 20.49) = 20.86$$

$$F_{12} = 20.86 + 0.05 (25 - 20.86) = 21.074$$

$$F_{13} = 21.07 + 0.05 (26 - 21.07) = 21.32$$

$$F_{14} = 21.32 + 0.05 (27 - 21.32) = 21.60$$

$$F_{15} = 21.60 + 0.05 (23 - 21.60) = 21.67$$

$$F_{16} = 21.67 + 0.05 (27 - 21.67) = 21.94$$

$$\alpha = 0.5$$

$$F_1 = 20 \quad F_2 = 20 + 0.5 (20 - 20) = 20$$

$$F_3 = 20 + 0.5 (21 - 20) = 23.5 \quad F_4 = 24.25$$

$$F_5 = 23.13 \quad F_6 = 20.56$$

$$F_7 = 20.78 \quad F_8 = 23.39$$

$$F_{10} = 18.6$$

$$F_{12} = 24.15$$

$$F_{14} = 26.04$$

$$F_{13} = 25.08$$

$$F_{15} = 24.52$$

$$F_{16} = 24.52 + 0.5 (27 - 24.52) = 25.7$$

Calculate

error

$$A_6 - F_6$$

DAY	1	2	3	4
M	43	51	40	64
T	45	41	57	58
W	22	31	30	33
T	25	22	33	38
F	31	25	37	25

1) Compute 3 days moving average

2) Plot the series and moving average on a graph

3) Does they appear to be a seasonal (weekly pattern)

2) 39, 37, 61, 58, 18, 52, 82, 27, 41, 69, 49, 66, 54, 42,

Use single exponential smoothing method with $\alpha = 0.2$

Ans)	DAY	①	MA	②	MA	③	MA
1)	M	43	-	51	-	40	-
	T	45	36.67	41	43	51	42.33
	W	22	30.67	37	33.33	30	40
	T	25	26	22	28	33	33.33
	F	31	-	25	-	31	-

2) Exponential Smoothing $F_{t+1} = F_t + \alpha (A_t - F_t)$

$$F_1 = 39$$

$$F_2 = F_1 + \alpha (A_1 - F_1) = 39 + 0.2 (39 - 39) = 39$$

$$F_3 = 39 + 0.2 (37 - 39) = 38.6$$

$$F_4 = 43.08$$

$$F_5 = 46.064$$

$$F_6 = 40.4512$$

$$F_7 = 42.76096$$

$$F_8 = 50.60877$$

$$F_9 = 45.88201$$

$$F_{10} = 44.90961$$

$$F_{11} = 49.72769$$

$$F_{12} = 49.58215$$

$$F_{13} = 52.86572$$

$$F_{14} = 53.09258$$

③ F_{t+1} = next period forecast

F_t = current year forecast

A_t = Actual Value

α = Smoothing constant $0 < \alpha < 1$

$$F_{t+1} = F_t + \alpha (A_t - F_t)$$

(or) $F_t = F_{t-1} + \alpha (A_{t-1} - F_{t-1})$

$$F_{t+1} = \alpha A_t + (1 - \alpha) F_t$$

$\alpha = 0.8$ Forecast reacts quickly to changes (Rapid response)
 0.2 - Slow & Smooth

$$\alpha = 0.5$$

Jun actual = $A_{\text{June}} = 130$

$F_{\text{June}} = 115$

$$\begin{aligned} F_{\text{July}} &= F_{\text{June}} + \alpha (A_{\text{June}} - F_{\text{June}}) \\ &= 115 + 0.5 (130 - 115) \end{aligned}$$

Calculate F_{July}

$$F_{\text{July}} = F_{\text{June}} + 0.5 (A_{\text{June}} - F_{\text{June}})$$

Jan	1100	A_1
Feb	120	A_2
Mar	130	A_3
Apr	125	A_4
May	140	A_5

$$F_{Jan} = 100$$

$$F_{Feb} = 100 + 0.2 (100 - 100) = 100$$

$$F_{Mar} = 100 + 0.2 (120 - 100) = 104$$

$$F_{Apr} = 104 + 0.2 (130 - 104) = 109.2$$

$$F_{May} = 109.2 + 0.2 (125 - 109.2) = 112.36$$

$$F_{June} = 112.36 + 0.2 (140 - 112.36) = 117.888$$

$$\text{Oct actual } A_{Sep} = 130$$

$$F_{Sep} = 115$$

Calculate

$$F_{Oct}$$

$$F_{Oct} = F_{Sep} + 0.5 (A_{Sep} - F_{Sep})$$

$$= 122.5$$

$$\begin{cases} Q_1 & 4.8 \\ Q_2 & 4.1 \\ Q_3 & 6 \\ Q_4 & 6.5 \end{cases}$$

$$\begin{cases} Q_1 & 5.8 \\ Q_2 & 5.2 \\ Q_3 & 6.8 \\ Q_4 & 7.4 \end{cases}$$

$$T_A = 6.4125$$

$$Q_1 = 5.725$$

$$Q_2 = 5.2$$

$$Q_3 = 7.025$$

$$Q_4 = 7.525$$

$$\begin{cases} Q_1 & 5.6 \\ Q_2 & 7.5 \\ Q_3 & 7.8 \\ Q_4 & 7.8 \end{cases}$$

$$\begin{cases} Q_1 & 6.3 \\ Q_2 & 5.9 \\ Q_3 & 8 \\ Q_4 & 8.4 \end{cases}$$

$$T_A = 6.38125$$

$$Q = 5.725$$

$$Q_2 = 5.2$$

$$SI = \frac{QA}{TA}$$

89

$$Q_3 = 7.015$$

$$Q_4 = 7.525$$

$$SI_1 = 0.8973$$

$$SI_2 = 0.830$$

$$SI_3 = 0.8109$$

$$SI_3 = SI_1 = 0.8973$$

$$SI_2 = 0.815$$

$$SI_3 = 1.1089$$

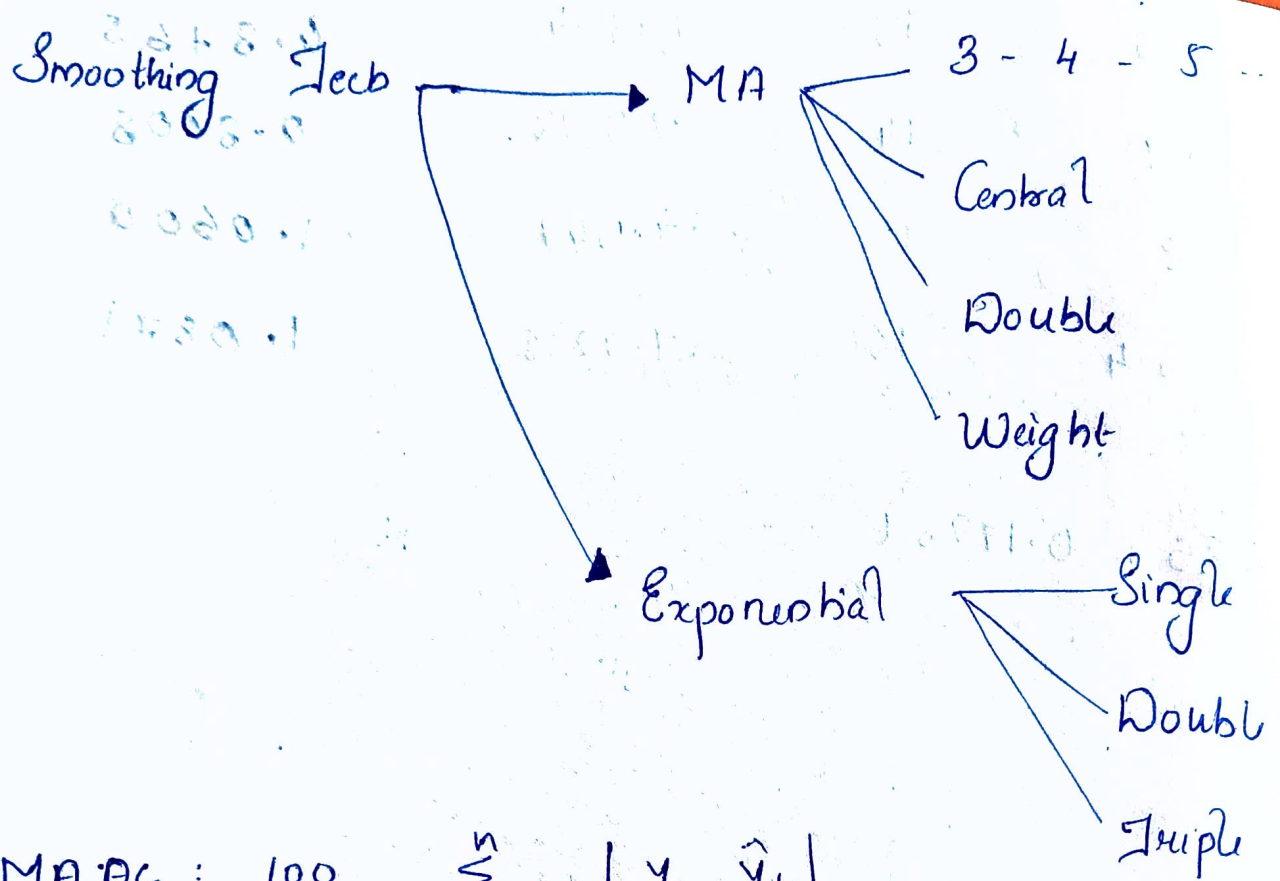
$$SI_4 = 1.179$$

Quarter	<u>4</u>	<u>6</u>	<u>7</u>	<u>7/7</u>	<u>SI</u>
Q ₁	4.8	1	5.0298	0.954	0.897
Q ₂	4.1	2	5.2096	0.787	0.815
Q ₃	6	3	5.3894	1.113	0.108
Q ₄	6.5	4	5.5692	1.167	1.179
Q ₁	5.8	5	5.749	1.009	0.89
Q ₂	5.2	6	5.9288	0.8770	0.81
Q ₃	6.8	7	6.1086	1.1132	0.108
Q ₄	7.4	8	6.2884	1.176	1.179
Q ₁	6	9	6.4682	0.927	0.89
Q ₂	5.6	10	6.648	0.8423	0.81
Q ₃	7.5	11	6.8278	1.098	0.1089
Q ₄	7.8	12	7.0076	1.1130	1.179

Q_1	6.3	13	7.1874	0.8765	0.89
Q_2	5.9	14	7.3672	0.8008	0.81
Q_3	8	15	7.547	1.0600	0.10
Q_4	8.4	16	7.7268	1.0871	1.11

$$J = 4.85 + 0.1798t$$





$$MAPE : \frac{100}{n} \sum_{t=1}^n \frac{|y_t - \hat{y}_t|}{\hat{y}_t}$$

$$MAD : \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t|$$

$$MSD : \frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2$$

MAPE - Mean Absolute Percentage Error

MAD - Mean Absolute Deviation

MSD - Mean Squared Deviation

$$F_{t+1} = \alpha (A_t - \hat{A}_t) + 0.5 \leq 1$$

Single :

$$S_t = \alpha A_{t-1} + (1-\alpha) (A_{t-1} + T_{t-1})$$

Double :

$$T_t = b(A_{t-1} - S_{t-1}) + (1-b) T_{t-1}$$

$$F_{t+1} = S_t + T_t$$

Time	Year	Sales	Smoothing $\alpha = 0.8$	Trend $\beta = 0.4$	F
0	1	2010	1451	0	1451
1	2	2011	1499	15.36	1504.76
2	3	2012	1686	73.35	1723.10
3	4	2013	1764	86.44	1842.26
4	5	2014	1819	98.19	1969.85
5	6	2015	1948	91.20	2043.51
6	7	2016	2013	81.42	2100.53
7	8	2017	2090	78.04	2170.15
8	9	2018	2173	78.95	2251.39
9	10	2019	2286	90.03	2369.13

Single = $F_{t+1} = F_t + \alpha(A_t - F_t)$
 Double = $F_t = aA_t + (1-a)(S_{t-1} + T_{t-1})$
 $T_t = b(S_t - S_{t-1}) + (1-b)T_{t-1}$
 $F_t = F_t + T_t$

A) H-W addition method

$$L_t = \alpha (y_t - S_{t-s}) + (1-\alpha) (L_{t-1} + b_{t-1})$$

$$b_t = \beta (L_t - L_{t-1}) + (1-\beta) b_{t-1}$$

$$S_t = \gamma (y_t - L_t) + (1-\gamma) S_{t-s}$$

$$F_{t+m} = L_t + m b_t + S_{t-s+m}$$

Exponential smoothing has three components

Single, Trend

1) Levels - It gives the current value

$$L_t = \alpha \frac{y_t}{S_{t-s}} + (1-\alpha) (L_{t-1} + b_{t-1})$$

eg: Current sales = 500

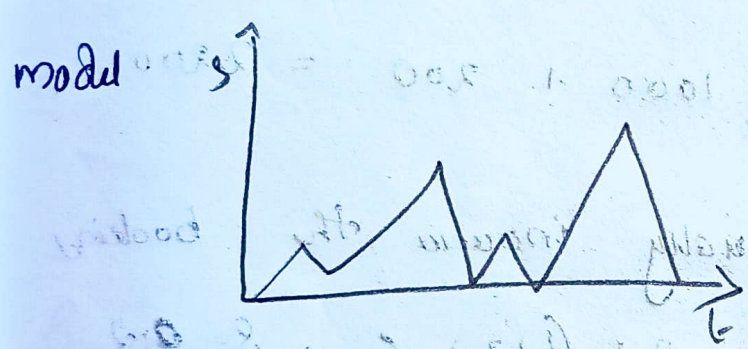
2) Trend - Are the sales going up or coming down. eg: Sales increases 20 unit by every month.

$$b_t = \beta \left(\frac{L_t}{L_{t-1}} \right) + (1-\beta) b_{t-1}$$

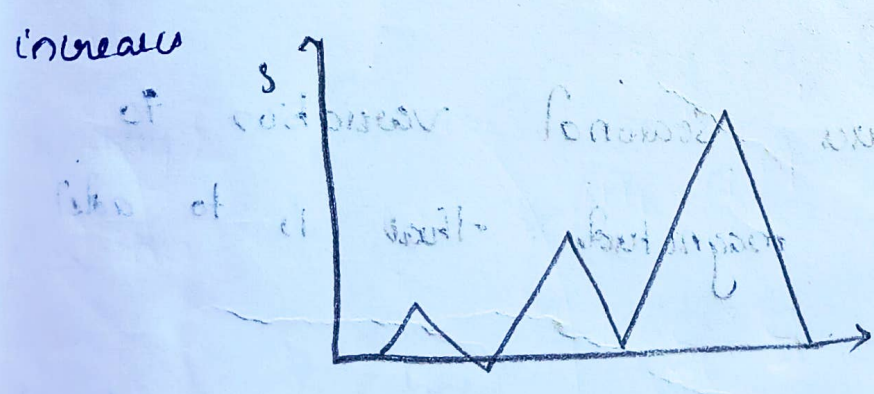
3) Seasonality - Repeated pattern

eg: December sales are of every year are usually higher than other.

Seasonal variation is same in the additive model



Seasonal variation becomes large as the time increases



Comparison of H.W. in

Comparative Comparison of additive & multiplicative model of in H.W model

Additive

Seasonal effect is same It changes with effect

Levels + Seasonality

Problems in Multiplicative H.W. Technique

1) Suppose a hotel has the current level 1000 bookings. Trend is +1000. December seasonal effect is +200. Forecast will be $L + T + S$.

$$L + T + S = 1000 + 1000 + 200 = 2200$$

2) Assume Dec month usually increases the booking by 20%. So the forecasting model $(L + T) \times S$. $S = 1.2$

$$(L + T) S = (1000 + 1000) \times 1.2 = 2400$$

Note: Pole printer uses seasonal variation to be roughly constant in magnitude. that is to add the same amount.

In multiplicative seasonal variation changes proportionally with levels of the time series data that is multiplied by same percentage.

Note: Monthly and quarterly data (given), calculate the seasonality and then de-seasonalise it. The multiplicative seasonality index is also.

B) H-W Multiplicative

$$L_t = \alpha \frac{y_t}{S_{t-s}} + (1-\alpha) (L_{t-1} + b_{t-1})$$

$$b_t = \beta \left(\frac{L_t}{L_{t-1}} \right) + (1-\beta) b_{t-1}$$

$$S_t = \gamma \left(\frac{y_t}{L_t} \right) + (1-\gamma) S_{t-s}$$

$$F_{t+m} = (L_t + m b_t) S_{t-s+m}$$

simple
Seasonality
Trend
Holt method
Holt method

Summary

Single : $F_{t+1} = F_t + \alpha (A_t - F_t)$

Double : $S_t = a A_t + (1-a) (S_{t-1} + T_{t-1})$ $0 \leq a \leq 1$

$T_t = b (S_t - S_{t-1}) + (1-b) T_{t-1}$ $0 \leq b \leq 1$

$F_{t+1} = S_t + T_t$

Triple : Addition H.W

$$L_t = \alpha (y_t - S_{t-s}) + (1-\alpha) (L_{t-1} + b_{t-1})$$

$$b_t = \beta (L_t - L_{t-1}) + (1-\beta) b_{t-1}$$

$$S_t = \gamma (y_t - L_t) + (1-\gamma) S_{t-s}$$

$$F_{t+m} = L_t + m b_t + S_{t-s+m}$$

1.1.1 Multiplication

$$L_t = \alpha \frac{y_t}{S_{t-1}} + (1-\alpha) (L_{t-1} + b_{t-1})$$

$$b_t = \beta \left(\frac{L_t}{L_{t-1}} \right) + (1-\beta) b_{t-1}$$

$$S_t = \gamma \left(\frac{y_t}{L_t} \right) + (1-\delta) S_{t-1}$$

$$F_{t+m} = (L_t - mb_t) S_{t+m}$$

ESM $F_{t+1} = F_t + d(P_t - F_t)$ $d = 0.5$

1) Q_1	Q_2	Q_3	Q_4	
2011	3.1	4.1	3.3	3.5
2012	3.1	3.9	3.6	3.6
2013	4	4.1	3.3	3.1
2014	3.3	4.4	4	4

$$QA_1 = 3.675$$

$$QA_3 = 3.55$$

$$QA_2 = 4.275$$

$$QA_4 = 3.55$$

$$TA = 3.7625$$

	Sales	SI	Deconsol
Q ₁	3.7	0.97	3.78
Q ₂	4.1	1.13	4.13
Q ₃	3.3	0.94	3.49
Q ₄	3.5	0.94	3.18
Q ₁	3.7	0.97	3.43
Q ₂	3.9	1.13	

	Sales	SI	Deconsol
Q ₁	3.7	0.97	3.78
Q ₂	4.1	1.13	4.13
Q ₃	3.3	0.94	3.49
Q ₄	3.5	0.94	3.18
Q ₁	3.7	0.97	3.43
Q ₂	3.9	1.13	
Q ₃	3.6	0.94	3.81
Q ₄	3.6	0.94	3.81
Q ₁	4	0.97	4.09
Q ₂	4.1	1.13	3.60
Q ₃	3.3	0.94	3.49
Q ₄	3.1	0.94	3.28
Q ₁	3.3	0.97	3.37
Q ₂	4.4	1.13	3.87
Q ₃	4	0.94	4.23
Q ₄	4	0.94	4.23

Deconsol
Sales/ST

7) GSM

$$F_{t+1} = F_t + \alpha(F_t - F_{t-1})$$

1.2
0.9
1
0.7
1.2
1.08

1.2
1.2
1.05
1.025
0.86
1.08

$$\alpha = 0.5$$

1.2
1.2
0.9
1
0.7
1.3

$$\alpha = 1$$

2) Double S

T

F12

E

64
82
92
95
106

64
73
84.75
93.93
104.298

0
4.5
8.125
8.656
9.507

64
77.5
92.875
102.586
113.805

0
4.5
-0.875
-7.586
-7.805

$$\alpha = 0.5$$

$$\beta = 0.5$$

$$S_t = \alpha F_t + (1-\alpha)(S_{t-1} + T_{t-1})$$

$$T_t = b(S_t - S_{t-1}) + (1-b)T_{t-1}$$

MA	MA (3)	DMA	CMA
3.1	-	-	-
4.1	3.9	-	3.865
3.3	3.83	3.743	3.665
3.5	3.5	3.616	-
3.7	3.7	3.643	3.6
3.9	3.73	3.71	3.715
3.6	3.7	3.72	3.715
3.6	3.73	3.776	3.715
4.0	3.9	3.81	3.815
4.1	3.818	3.73	3.85
3.3	3.548	3.51	3.65
3.1	3.23	3.443	3.365
3.3	3.6	3.516	3.415
4.4	3.9	-	3.75
4.0	-	-	-

Seasonality $\rightarrow Q$
Trend $\rightarrow$ Regression

Q. 48

Q 54

Q. 6

Q. 41

Q. 63

Q. 54

Q. 68

A. 64

Q. 78

Q. 61

690

Q. 18

10 6 15

Q. 53

Q₂ 6-93

Q. 7-23

57. 11/21/48

Cr. 1-1072

4470

பெயர்: _____ இனம்: _____ இயல்பு: _____

ကျေးဇူးတင်စရာပါသလား? 299

226 00

21400

238 60

82

242

圖書

143

3496

y - 3.194

Each square marked (circled by

Journal of Management Education

ESM	MA(3)	DMA	CMA	Holt (Double)
10				3
8	8		8.665	5
9	9.33	9.111	9.665	9
11	10			10
10				12
				11

$$F_{t+1} = y_t + \alpha (D_t - F_t)$$

$$\alpha = 0.5$$

$$S_t = \alpha A_t + (1-\alpha)(S_{t-1} + T_{t-1})$$

$$T_t = b(S_t - S_{t-1}) + (1-b)T_{t-1}$$

at model should be chosen

$$F_t = S_t + T_t$$

by considering the least error method.

is found by plugging the hands in

$$\text{Sales} = -255x + 2228$$

equation.

Errors

$$\text{MAPE} = \frac{100}{n} \sum_{t=1}^n \frac{|y_t - \hat{y}_t|}{\hat{y}_t}$$

$$\text{MAD} = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t|$$

$$\text{MSD} = \frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2$$

ESM

10

8

9

11

10

$$F_{t+1} = 10 + 0.5 \times (10 - 10) = 10$$

$$F_{\text{June}} = F_{\text{May}} + 0.5 (A_{\text{May}} - F_{\text{May}})$$

$$A_{\text{May}} = 100$$

$$F_{\text{May}} = 130$$

$$F_{\text{June}} = 130 + 0.5 (100 - 130) = 115$$

Holt's (3) (S_t) T_t

F_t

$$3 \quad (3.25) \quad 0.5$$

3

$$5 \quad 4.5 \quad 0.5$$

4.5

$$9 \quad 6.75$$

$$1.625$$

$$8.375$$

$$10 \quad 9.1875$$

$$2.03125$$

$$11.21$$

$$12 \quad 11.60$$

$$2.22$$

$$13.835$$

$$11 \quad 15.41$$

$$3.017$$

$$18.435$$

$$\alpha = \beta = 0.5$$

H W method is used for forecasting T.S that contains levels, trend & Seasonal Pattern to predict future values.

HW = current sales + trend ($\uparrow / \downarrow$) + seasonal effect / pattern

Retail - Hotel - Education - Airlines - Electricity - Hospitals

E commerce

Banking

Additive :- Seasonal effect remains the same constant
(add S) [100 units every year (+100)]

Multiplicative :- Seasonal effect changes ($\uparrow/\downarrow$) with
(* S) overall levels (20% high than normal)

If actual sales is 1000

Sales in Dec 1200

Levels

Current situation

(current value of data)

Current sales is 500

100 - 200 - 3000 - 4000

Where am I now?

α controls (how $\times$)

γ gives for relevant observation)

Trends

Data $\uparrow/\downarrow$

Sales $\rightarrow$ 1000

which direction?

Where am I now?

β how quickly
the estimate bend
changes

Seasonality

Repeated pattern over
time

Sales $\rightarrow$ 2003

2004

2005

Does the pattern
Repeat?

if $\rightarrow$ how quickly
the seasonal patterns
changes

$$L_t = \alpha (y_t - S_{t-m}) + (1-\alpha) (L_{t-1} + T_{t-1})$$

$$T_t = \beta (L_{t-m} - L_{t-1}) + (1-\beta) T_{t-1}$$

$$S_t = \delta (y_t - L_t) + (1-\delta) S_{t-m}$$

h (forecasting period)

$$F_{t+h} = L_t + hT_t + S_{t-m+h}$$

h (forecasting period)

m (seasons in a year)

l - times

α, β, γ - Some constants

$$0 \leq \alpha, \beta, \gamma \leq 1$$

T.S.

Trend

- Linear
- Quadratic
- Exponential

MA/ESM

S

D

T

Ex

Holt

H-W

(Trend no)

(TREND)

(T & S)

eq.

	Q_1	Q_2	Q_3	Q_4
2024	100	120	140	160
2025	120	140	160	180

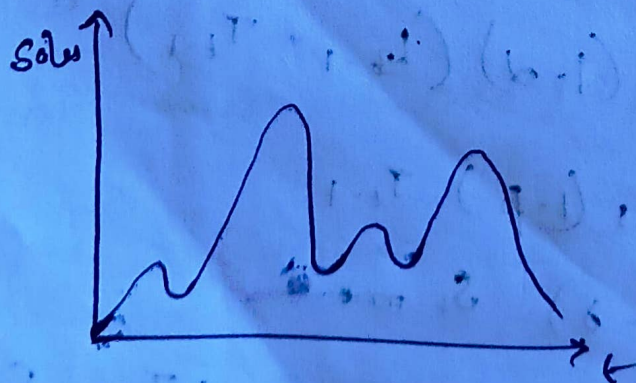
100

120

140

160

180



Seasonal Indices

Additive model : $AQ_i - TA$

Multiplicative model : AQ_i / TA

Ans. $TA = 140$

$$AQ_1 = 110$$

$$AQ_2 = 130$$

$$AQ_3 = 150$$

$$AQ_4 = 170$$

$$SI_1 = 110 - 140 = -30$$

$$SI_2 = 130 - 140 = -10$$

$$SI_3 = 150 - 140 = 10$$

$$SI_4 = 170 - 140 = 30$$

$$SI_1 = 110/140 = 0.78$$

$$SI_2 = 130/140 = 0.92$$

$$SI_3 = 150/140 = 1.071$$

$$SI_4 = 170/140 = 1.214$$

Additive

Multiplicative

Current data = 180 = L_8

for that $SI = 30 = S_8$

$$T = L_8 - (\text{overall average}) = 180 - 140 = 40$$

$$F_8 = T + S + L = 40 + 30 + 110 = 180$$

$$L_8 = 180 + 30 = 210 \text{ or } L_8 = 180 - 30 = 150$$

$$T = 210 - 140 = 70$$

$$T = 150 - 140 = 10$$

$$F_8 = T + S + L = 10 + 30 + 110 = 150$$

$$150 + 10 - 30 = 130 \checkmark$$

$$L_8 = 180$$

$$S_8 = 30$$

$$S_8 = -30$$

$$L_8 = L_8 - S_8$$

$$= 180 - 30 = 150$$

$$T_q = L_q - \text{Overall average} = 150 - 140 = 10$$

$$F_q = 150 + 10 - 30 = 130$$

H.W

$$\text{Current level} = 100$$

Trend

5

$$\text{Per-quarter Season} = 0.94 \text{ (H)}$$

$$L_t = 100$$

$$T_t = 5$$

$$S_t = 0.94$$

$$\text{Forecast} = (L_t + T_t) S_t = (100 + 5) \times 0.94 = 94.5$$

$$TA = 120$$

Y_1	100	120	110	136
Y_2	110	130	120	140

Additive

$$Q_1 = 105 - 120 = -15$$

$$Q_2 = 125 - 120 = +5$$

$$Q_3 = 115 - 120 = -5$$

$$Q_4 = 135 - 120 = +15$$

Multiplicative

$$\begin{aligned} SI_1 &= 105/120 = \\ SI_2 &= 125/120 = \\ SI_3 &= 115/120 = \\ SI_4 &= 135/120 = \end{aligned}$$

Q_1 Q_2 Q_3 Q_4

68

62

61

63

58

66

61

65

63

63

67

68

Multiplicative

$$SI_1 = \frac{Q_1}{TA} = \frac{67}{63.75}$$

$$SI_2 = \frac{Q_2}{TA} = \frac{61}{63.75}$$

$$SI_3 = \frac{Q_3}{TA} = \frac{63.3}{63.75}$$

$$SI_4 = \frac{Q_4}{TA} = \frac{63.67}{63.75}$$

Additive SI would be numbers

Multiplicative SI would be decimal point

Q) Current level = 100 $SI = 5$ Trend = 5

Find $Q_3 = 100 + (5 \times 3) + 5 = 120$

Seasonalise Actual / SI

SI $\left\{ \begin{array}{l} \text{Add: } Q_n - TA ; L + T + S \\ \text{Mult: } Q_n / TA ; (T+L)^* S \end{array} \right.$